

Khush Jain

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EDUCATION

University of Wisconsin–Madison

B.S. in Computer Science & Data Science; Minor in Geospatial Data Science

Madison, WI

Sep 2023 – May 2027

GPA: 3.6/4.0 (Dean's List)

EXPERIENCE

Undergraduate Researcher – Multimodal Foundation Models

Sep 2026 – Present

GeoDS Lab, University of Wisconsin–Madison; Advisor: Prof. Song Gao

Madison, WI

- Developing PyTorch pipelines for multimodal preprocessing, representation learning, and evaluation, benchmarking cross-modal alignment and geographic transfer across GeoAI tasks.

Software Engineering & Geospatial Computing Intern

Aug 2026 – Present

MultiSet AI

Madison, WI

- Implemented OGC GeoPose and Python REST APIs for 6-DoF metadata validation and coordinate transforms, reducing cross-platform pose-conversion latency by **35%**.
- Fused point clouds and computer vision to detect low-feature-density regions and predict localization drift, improving 3D localization accuracy by **25%**.

Software Engineering & DevOps Intern

Jun 2026 – Aug 2026

Mass General Brigham Digital Innovation

Boston, MA

- Built and deployed a GitHub Copilot analytics platform for **561 seats** using TypeScript, Vue/Nuxt, Node.js/Nitro, GitHub REST APIs, and PostgreSQL.
- Developed **10+ analytics modules** for executive KPIs, usage trends, cost attribution, and license efficiency across 244 developers and 144K+ chat interactions.
- Implemented GitHub OAuth, server-side RBAC, backend filtering, and tests; deployed Docker containers to OpenShift Kubernetes clusters and flagged **267 unused seats** (up to **\$8K/month** in potential waste).

Cloud & Artificial Intelligence Intern

May 2025 – Aug 2025

Mass General Brigham Digital Innovation

Boston, MA

- Developed an assistant using NLP and semantic search over **2,000+ clinical workflow and policy documents**, reducing repetitive support tickets by **28%**.
- Integrated intent classification and named-entity recognition with low-confidence escalation, achieving **91% routing accuracy** across 15 inquiry categories.
- Built a React/FastAPI interface serving cited LLM answers in **under 2.5 seconds** to 1,500+ internal pilot users.

PROJECTS

Distributed & Cloud Data Systems | *Python, gRPC, Spark, Kafka, Cassandra, HDFS*

Nov 2025

- Built 8 distributed systems spanning an 8-threaded gRPC service under a 512 MB memory limit, SQL-to-HDFS ETL, Spark, Cassandra, and Kafka; processed **447K+ loan records**.
- Engineered fault tolerance with replication, retries, consistency-level tuning, checkpoint recovery, and atomic HDFS writes; built a 4-partition Kafka pipeline with crash-safe Parquet storage.

CodeAtlas | *Python, FastAPI, FAISS, MCP, RAG*

Mar 2025

- Built a RAG assistant with semantic chunking, vector embeddings, and FAISS indexing across **1,000+ technical documents**, serving cited answers through FastAPI.
- Exposed schema-validated search, top-*k* retrieval, and citation tools through an MCP server for external AI clients.

TerraMind | *Python, React, PostGIS, GeoPandas, Leaflet*

May 2026

- Built a React/Leaflet map and Python APIs for spatial search over multi-source GeoJSON in PostGIS, with optimized large-dataset rendering.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, TypeScript, JavaScript, C#, C, SQL, Node.js, React, Vue/Nuxt, FastAPI

AI & ML: PyTorch, Transformers, RAG, FAISS, MCP, LoRA, Quantization, Prompt Engineering

Cloud & Systems: Azure, Docker, Kubernetes, OpenShift, Terraform, Linux, Git, CI/CD, Bash, Bicep

Data & Tools: PostgreSQL, MySQL, Spark, Kafka, Cassandra, HDFS, Pandas, GeoPandas, ArcGIS Pro